WQF7009 EXPLAINABLE AI

ALTERNATIVE ASSESMENT
EXPLAINABILITY OF LESION CLASSIFICATION MODEL

GROUP INFORMATION

- GROUP MEMBERS:
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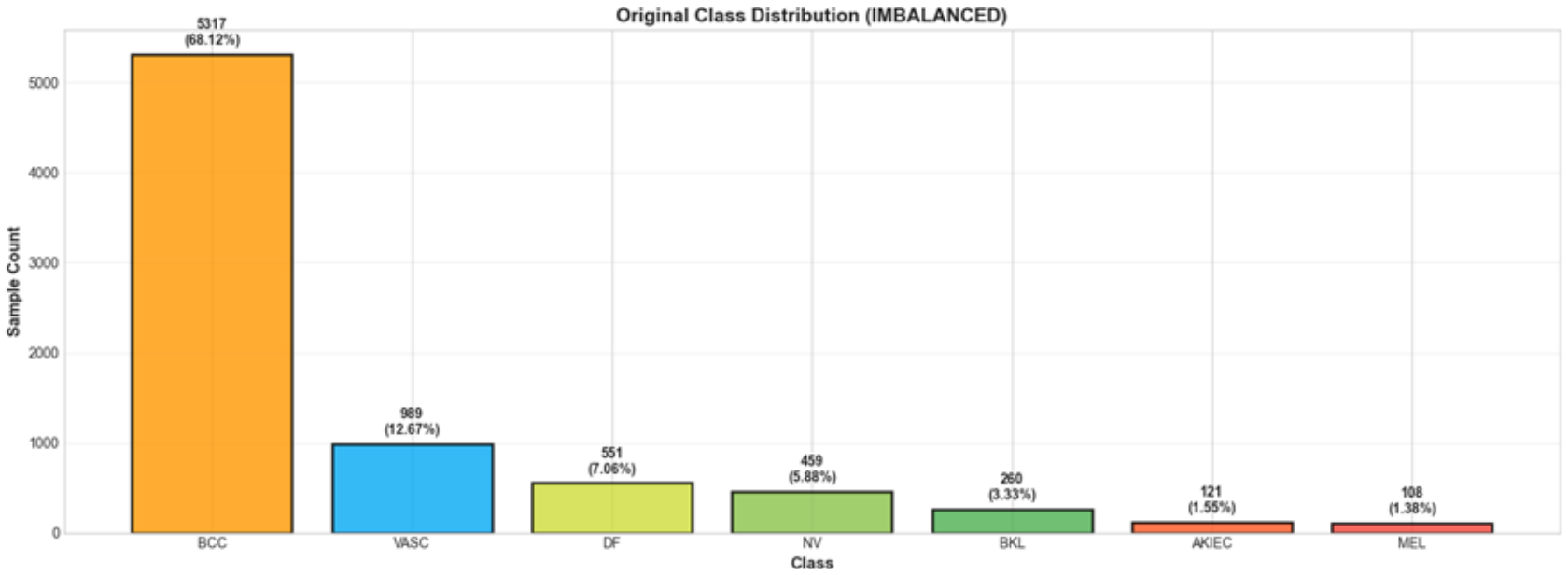
Introduction

- We present a deep learning–based skin lesion classification model designed to categorize dermoscopic images into seven clinically relevant classes
- GradCAM and feature ablation method as xAI methods

National Cancer Society Malaysia
Giving Hope Celebrating Life

Dataset

- The dataset used in this study was obtained from the Skin Lesion Segmentation and Classification collection available on Hugging Face.
- Contains 7 classes.
- 9,547 dermoscopic images in total
- The dataset is heavily imbalance

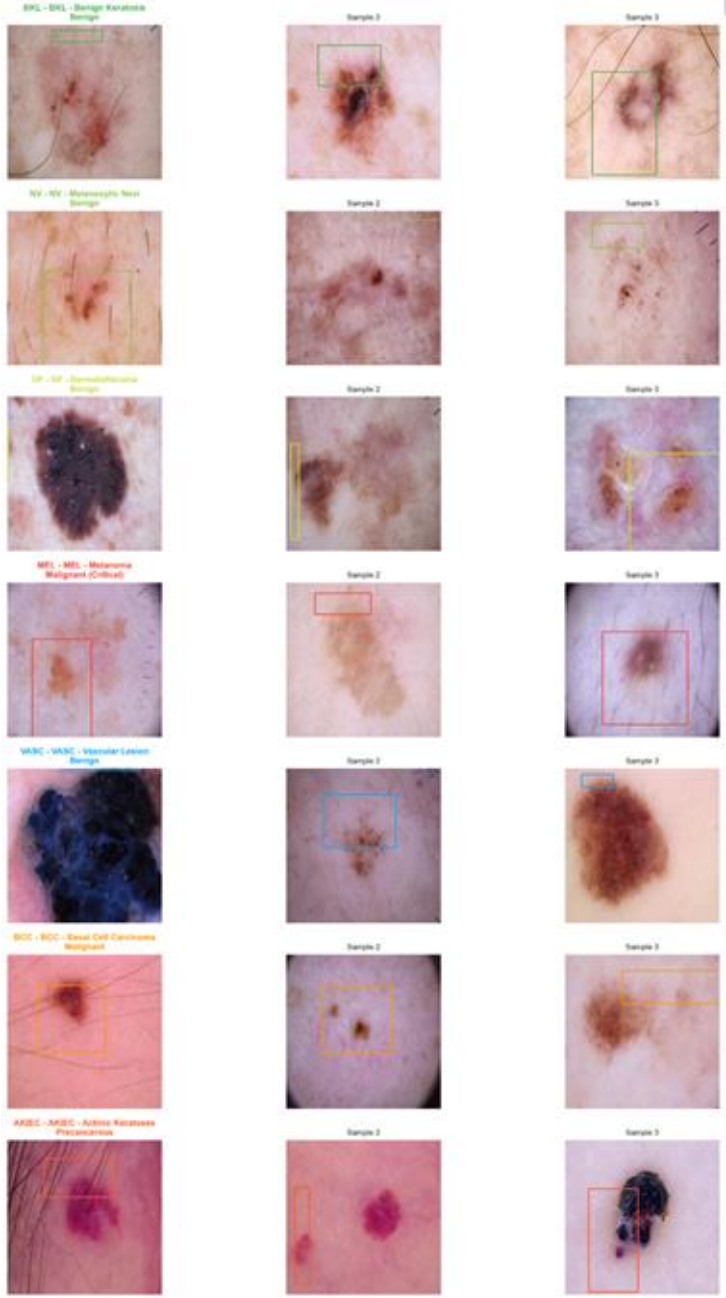


Stakeholder Questions

- **Clinicians**
- How confident is the model in this prediction?
-
- **Patients**
- Is this something I should worry about?
-
- **Data Scientists**
- Where does the model fail or show uncertainty?

Class ID	Abbreviation	Lesion Type	Clinical Description
0	BKL	Benign Keratosis	Non-cancerous, often scaly skin lesions. Common and usually harmless.
1	NV	Melanocytic Nevi	Regular moles formed by pigment-producing cells. Typically benign.
2	DF	Dermatofibroma	Firm, small nodules under the skin caused by minor trauma. Non-cancerous.
3	MEL	Melanoma	A serious and potentially life-threatening skin cancer. Early detection is critical.
4	VASC	Vascular Lesion	Blood vessel-related lesions such as angiomas; usually red or purple.
5	BCC	Basal Cell Carcinoma	The most common form of skin cancer; slow-growing and rarely metastasizes.
6	AKIEC	Actinic Keratoses	Pre-cancerous lesions that may progress to squamous cell carcinoma.

- The bounding boxes provided are not accurate.
- The same type of lesions can come in multiple colorations and sizes.
- Almost all of the patients have fair skin



Dataset Preprocessing



EDA & Gloss Selection

- Perform simple EDA



Cleaning & Inspection

- Ensure all image has its corresponding label



Dataset Balancing

- Use undersampling to balance the dataset

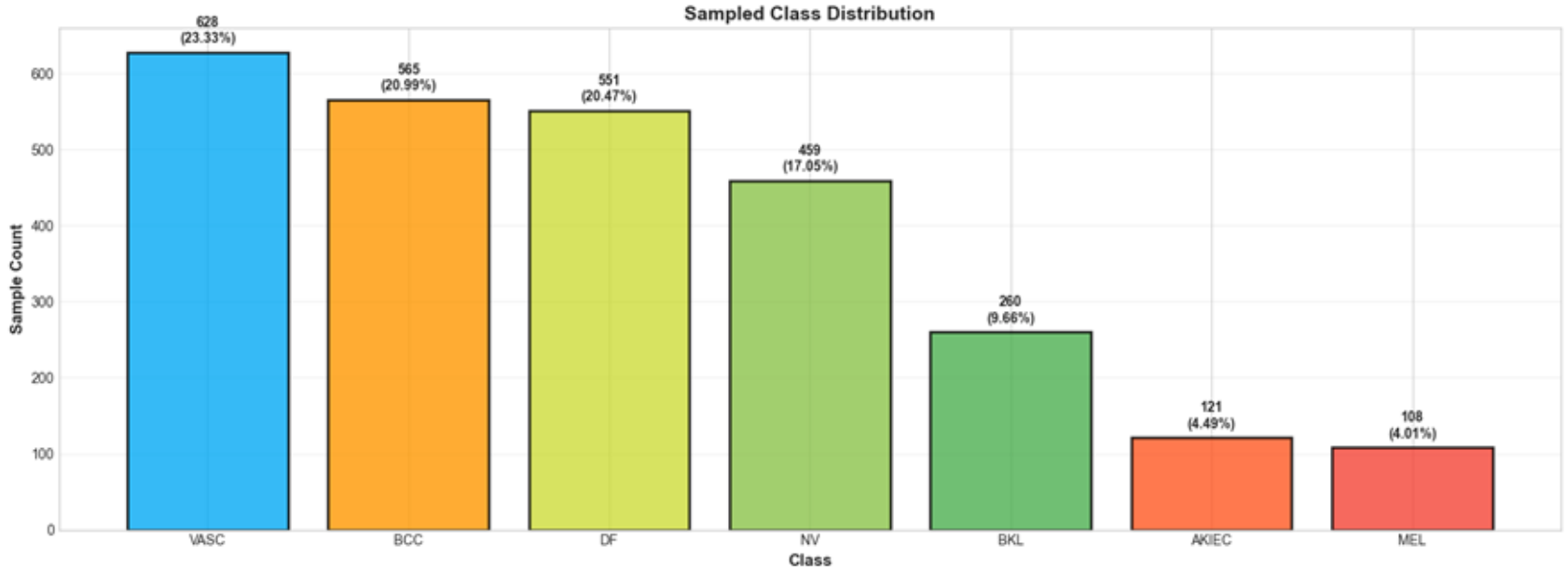


Storage

- Post processed dataset stored on Drive for model training

class_id	class_code	severity	count	percentage
4	VASC	Benign	628	23.33
5	BCC	Malignant	565	20.99
2	DF	Benign	551	20.47
1	NV	Benign	459	17.05
0	BKL	Benign	260	9.66
6	AKIEC	Precancerous	121	4.49
3	MEL	Malignant (Critical)	108	4.01

```
Imbalance Metrics:
Most common: VASC with 628 samples
Least common: MEL with 108 samples
Imbalance ratio: 5.81:1
```



Base Model - Overview

1. Configuration

- Finetune a ResNet18 Model
- Replace the final fully connected layer with a new linear classification head consisting of seven output neurons
- All convolutional layers of the pretrained backbone were frozen during training

2. Hyper parameter

- Early stopping ends at 12th epoch.
- Optimizer: ADAM
- Learning rate: 1x10^-3
- Loss function: Cross Entropy with Weighted Class

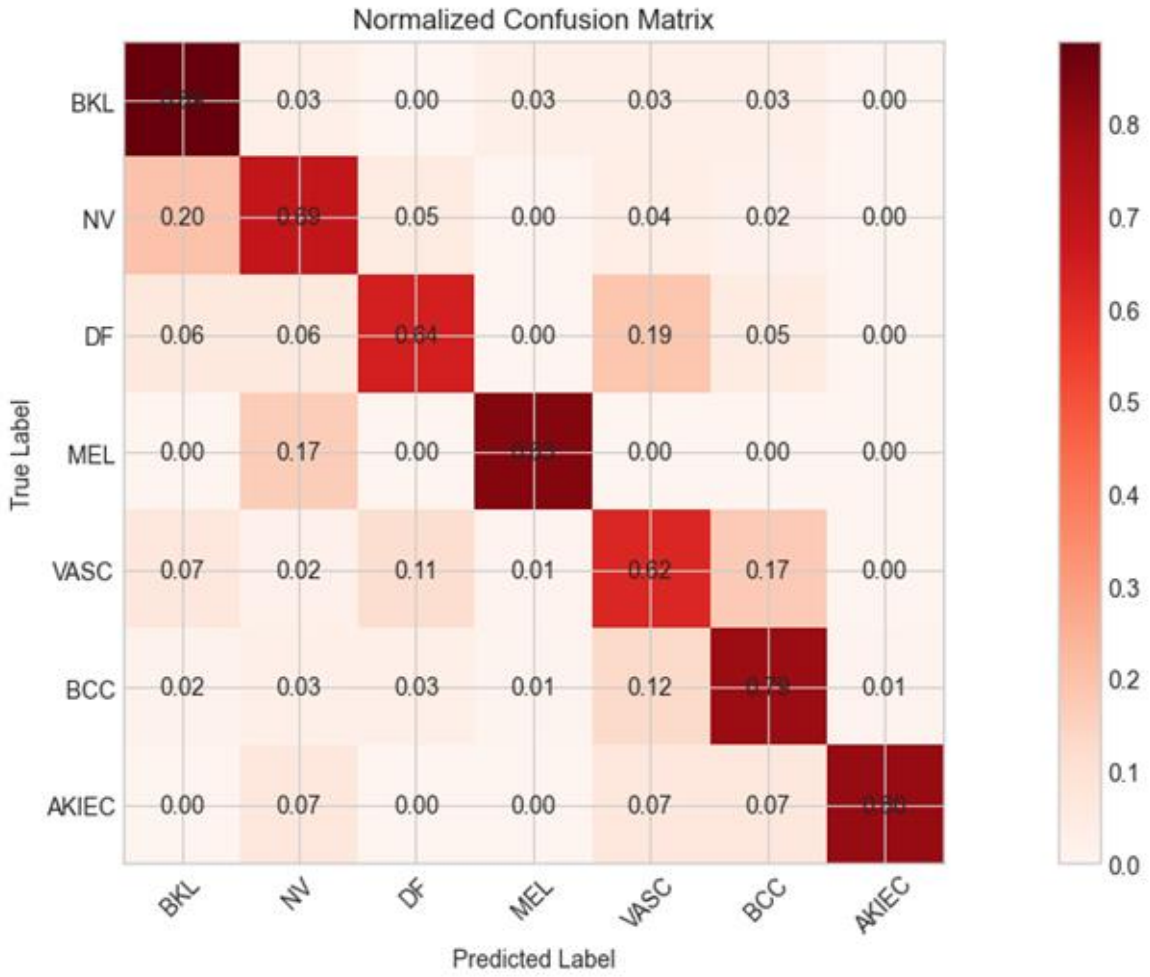
2. Evaluation Metric

- Weighted Accuracy, Recall and F1 Score

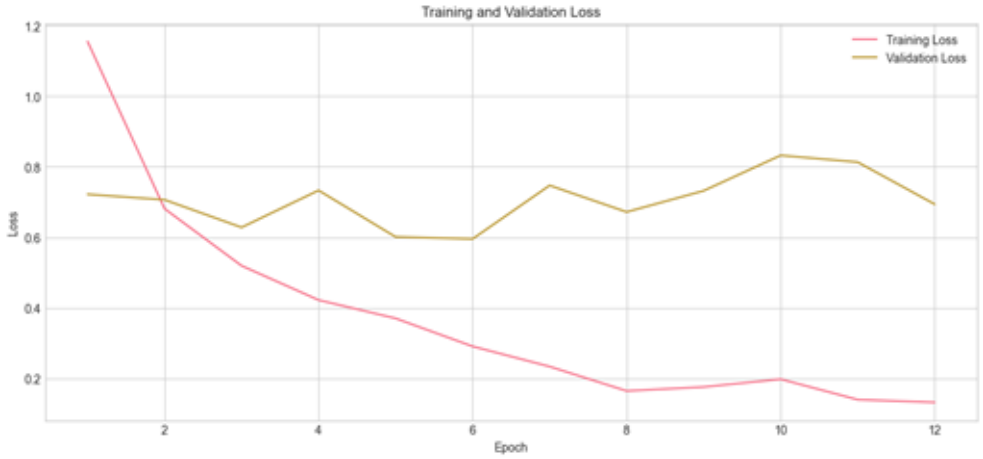
Preliminary Results

Weighted Accuracy	0.7596
Weighted Recall	0.7596
Weighted F1	0.7770

Class	Precision	Recall	F1	Support
0 (BKL)	0.492	0.889	0.634	36
1 (NV)	0.585	0.691	0.633	55
2 (DF)	0.539	0.641	0.586	64
3 (MEL)	0.625	0.833	0.714	12
4 (VASC)	0.418	0.617	0.498	115
5 (BCC)	0.953	0.792	0.865	664
6 (AKIEC)	0.706	0.800	0.750	15



Preliminary Results





Feature Importance xAI: Grad-CAM

Grad-CAM is architecture-agnostic

Given a CNN and an input image, Grad-CAM computes the importance of each feature map in a selected convolutional layer with respect to a target class c.

The output of the last convolutional block was used as the feature map, A^k for Grad-CAM computation.

Counter Factual xAI: Occlusion Method

Systematically perturbs the input image to observe changes in the model's confidence.

The implementation utilizes a sliding window approach.

A square patch of size $k\ x\ k$ is moved across the image with a specific stride.

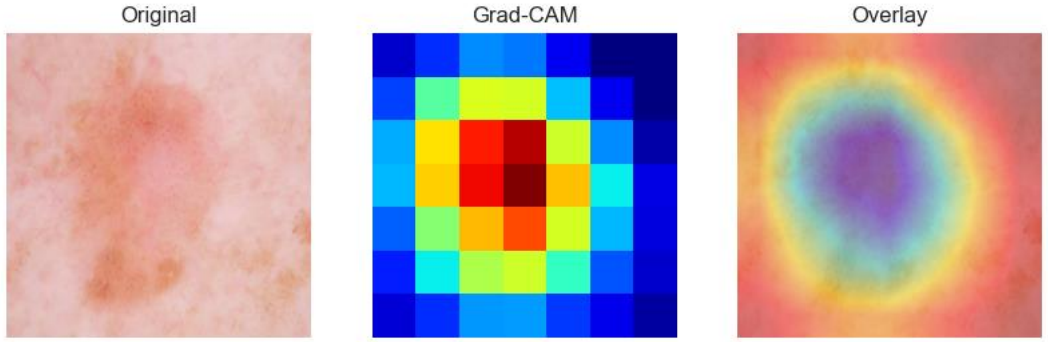
At each location (i, j), the underlying pixels are replaced with a baseline value.

The "importance" or sensitivity of the occluded region is calculated as:

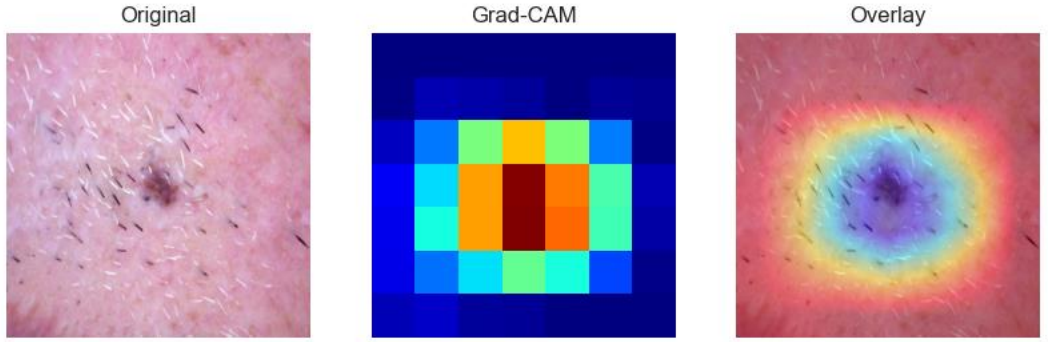
$$Importance_{i,j} = P(class \mid Image_{orig}) - P(class \mid Image_{occ})$$

Grad Cam Results

Class 0 : Benign Keratosis

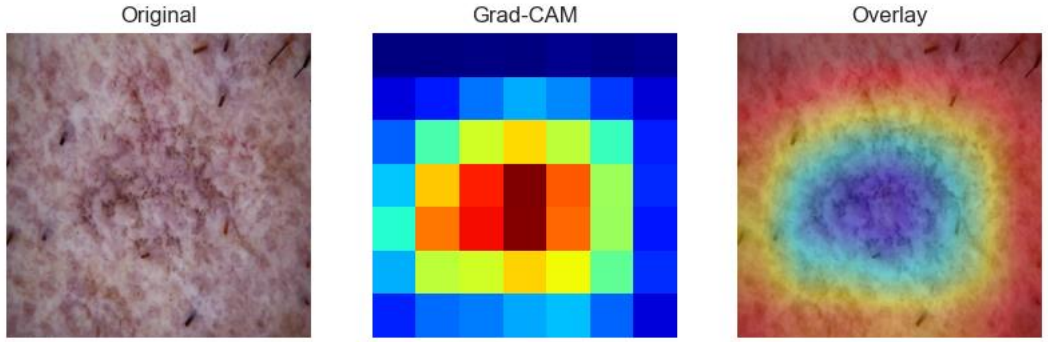


Class 1 : Melanocytic Nevi

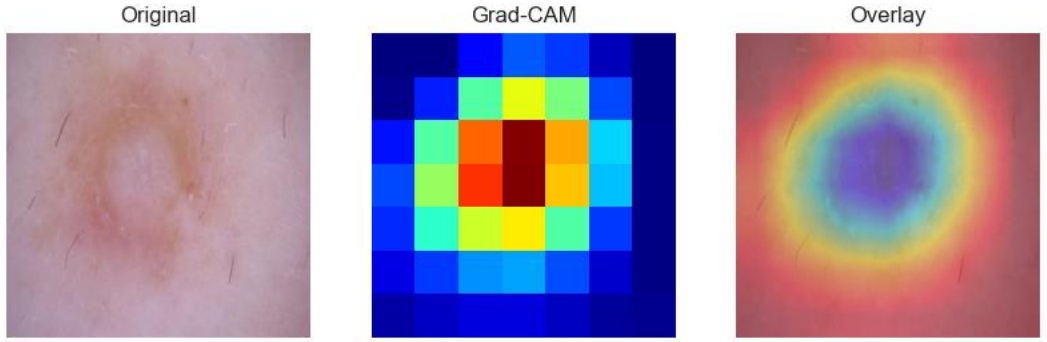


Grad Cam Results

Class 2 : Dermatofibroma

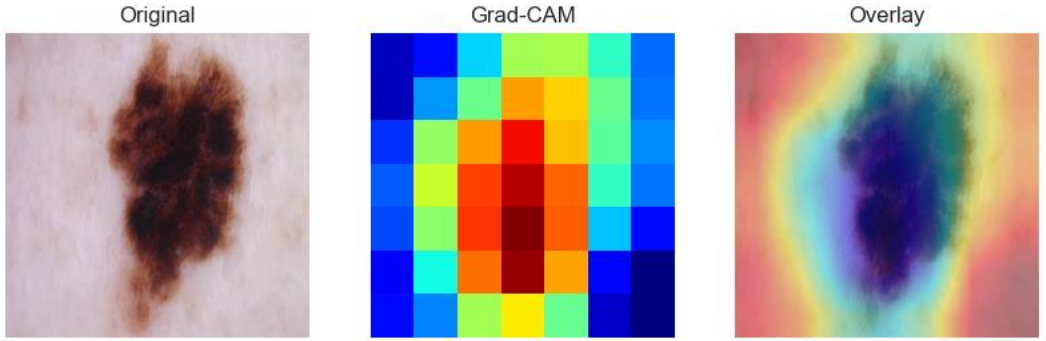


Class 3 : Melanoma

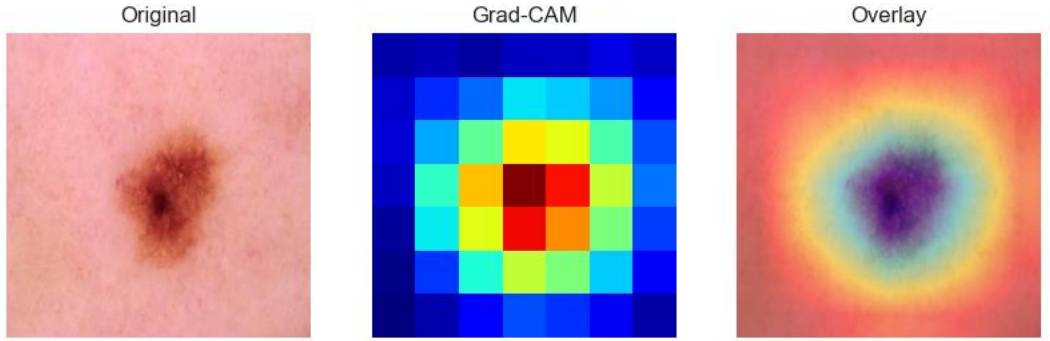


Grad Cam Results

Class 4 : Vascular Lesions

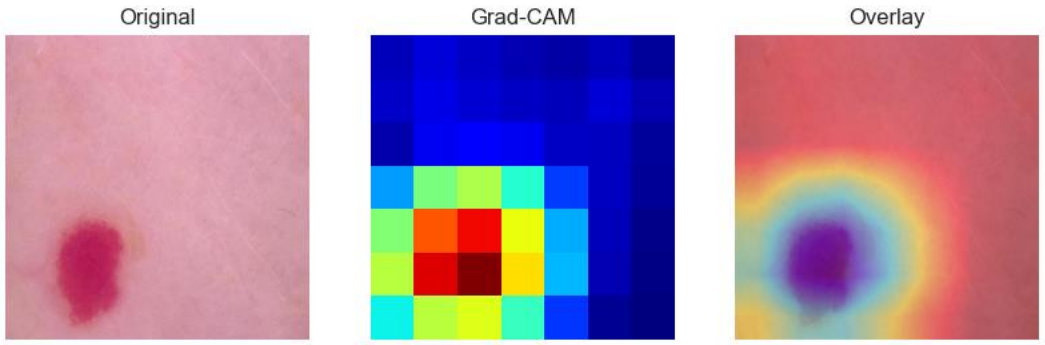


Class 5 : Basal Cell Carcinoma



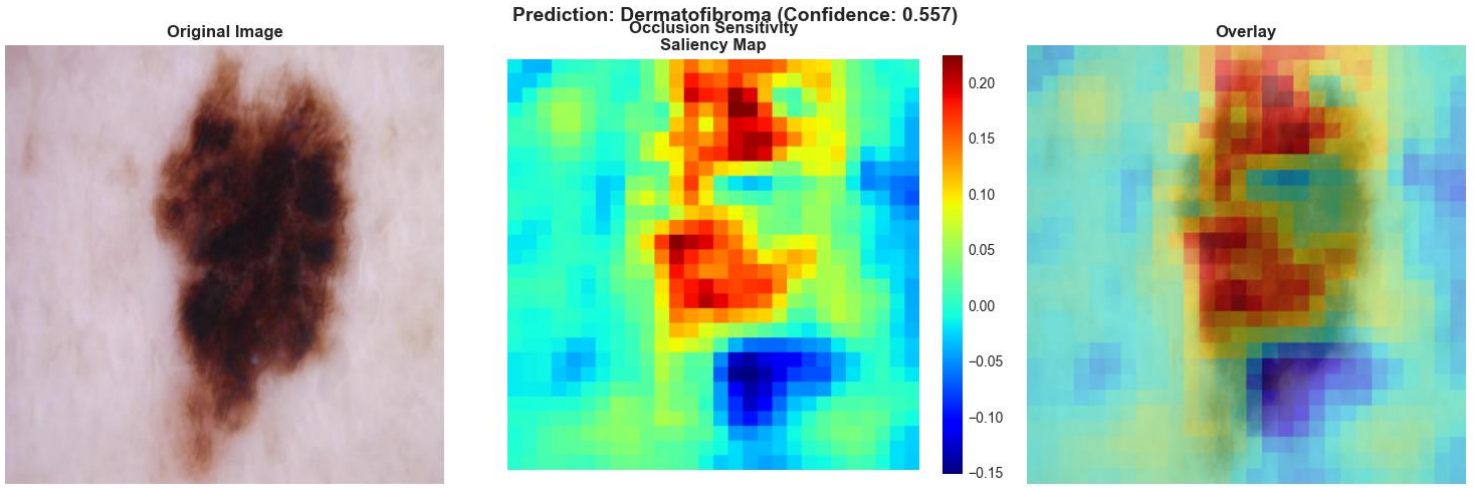
Grad Cam Results

Class 6 : Actinic Keratose



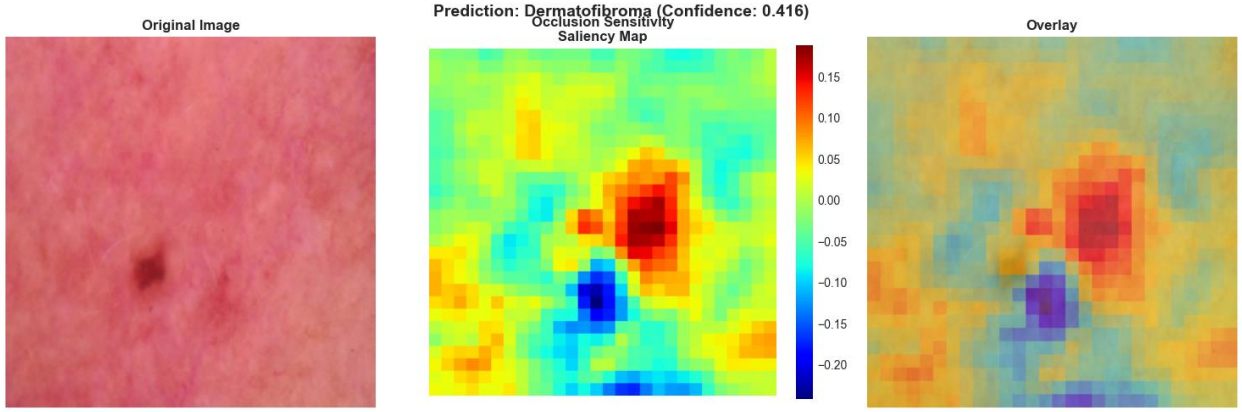
Occlusion Method Results

Correctly classified as benign keratosis by our model. However, after occlusion method is applied the model classified it as dermatofibroma.



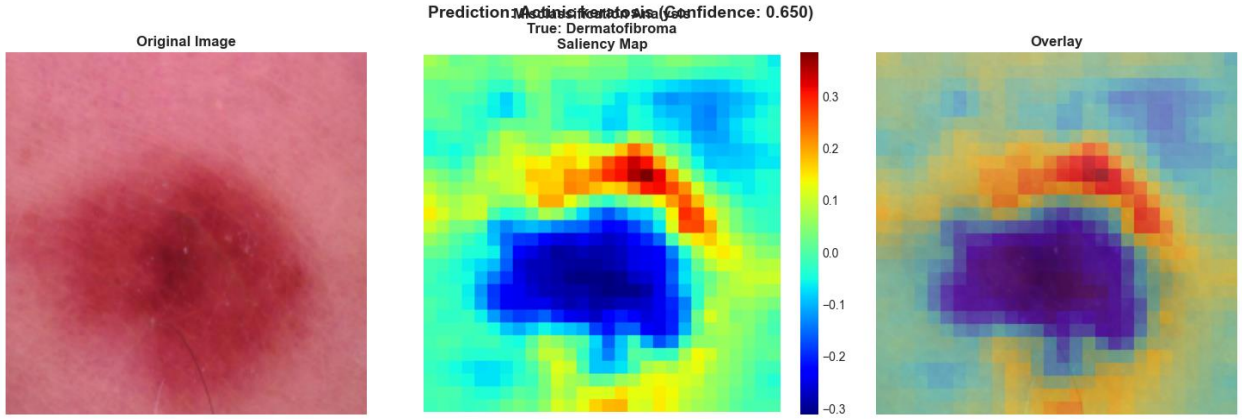
Occlusion Method Results

The model correctly classified it as dermatofibroma but there is a sharp drop in confidence.

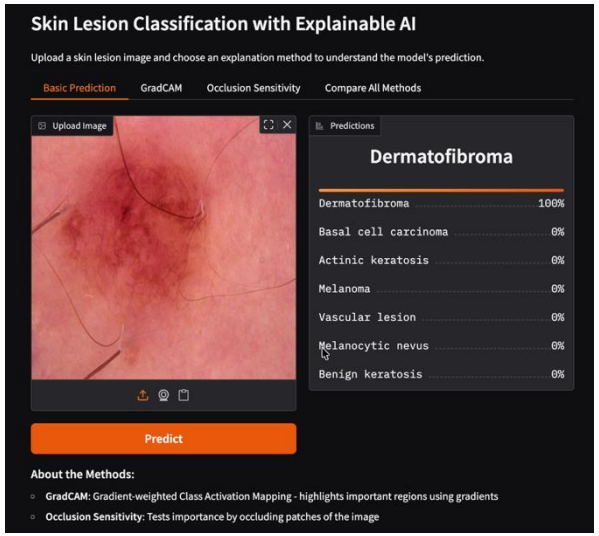


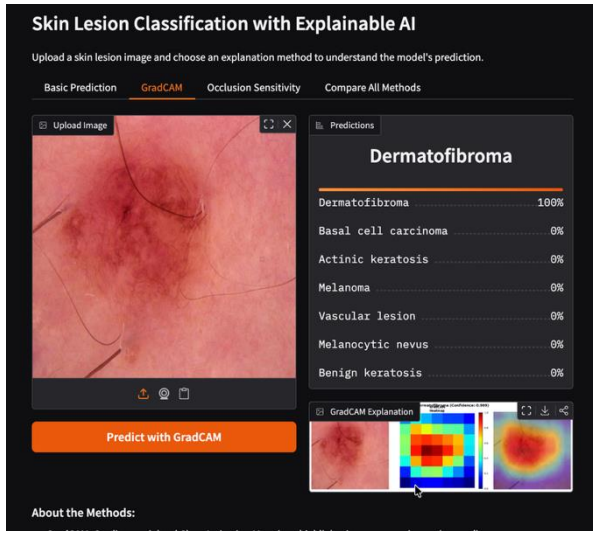
Occlusion Method Results

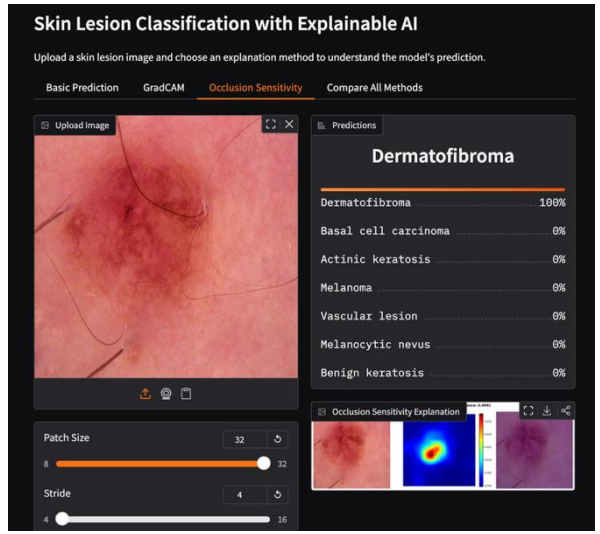
The model misclassified dermatofibroma as actinic keratosis after occlusion is applied.



Dashboard


Skin Lesion Classification with Explainable AI
Upload a skin lesion image and choose an explanation method to understand the model's prediction.
Basic Prediction
GradCAM
Occlusion Sensitivity
Compare All Methods
Upload Image
Predictions
Dermatofibroma
Dermatofibroma 100%
Basal cell carcinoma 0%
Actinic keratosis 0%
Melanoma 0%
Vascular lesion 0%
Melanocytic nevus 0%
Benign keratosis 0%
Predict
About the Methods:
GradCAM: Gradient-weighted Class Activation Mapping - highlights important regions using gradients
Occlusion Sensitivity: Tests importance by occluding patches of the image


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Benign keratosis 0%
GradCAM Explanation
Predict with GradCAM
About the Methods:


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Actinic keratosis 0%
Melanoma 0%
Vascular lesion 0%
Melanocytic nevus 0%
Benign keratosis 0%
Occlusion Sensitivity Explanation
Patch Size
32
8
32
Stride
4
4
16

Conclusion

The integration of XAI into skin lesion classification addresses critical barriers to AI adoption in dermatology:

Enhanced Trust

Educational Value

Error Detection

Regulatory Compliance

We proposed two additional steps for the future:

1. Conduct formal studies with practicing dermatologists to evaluate whether XAI explanations improve diagnostic accuracy, confidence, and decision-making efficiency.

2. A holistic comparison between the full results from our model and our counterfactual method.

Thank you